

Machine Translation and Attention Mechanism

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Machine Translation

- *One of the oldest, hardest, but extremely useful problem of NLP*

Source language (S)

Target language (T)

I saw a girl with a cat.

⇒ मैंने एक बिल्ली के साथ एक लड़की को देखा।

Saya melihat seorang gadis dengan kucing.

⇒ ?

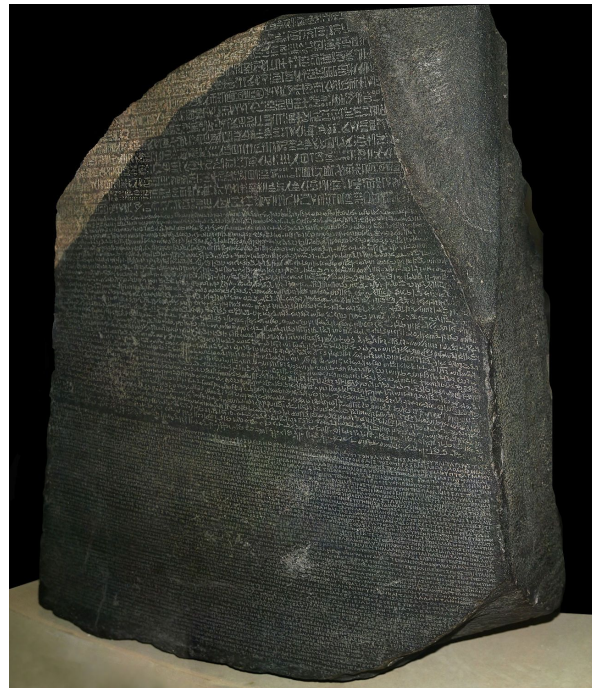
(Indonesian)

Major Challenges

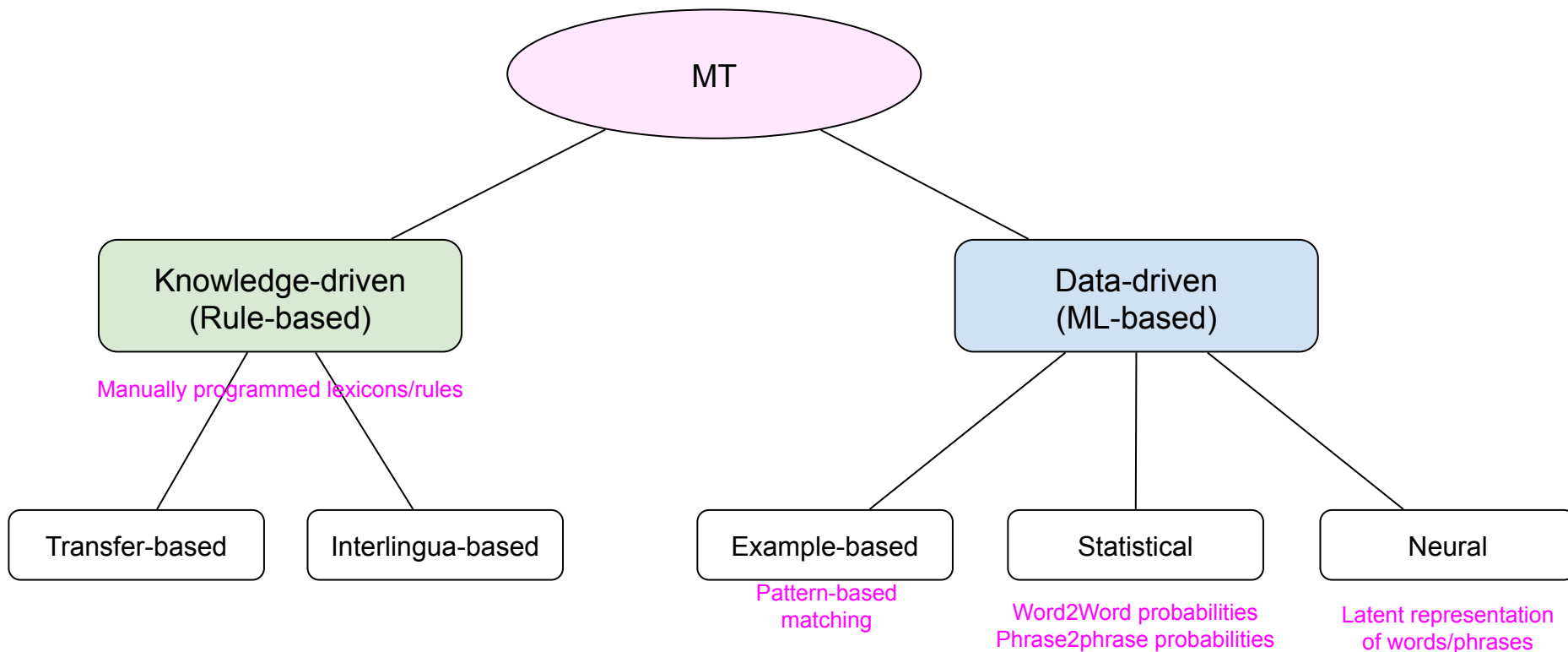
- Data
- Domain
- Incoherent sentences (Limited context in translation)
- Rare/Unknown words
- Lexical divergence
 - Syntax: SVO (English), SOV (Hindi), VSO, OSV
 - Free word order (Sanskrit) vs rigid word order (English)
 - Inflections: Infixing (Arabic), Agglutinative (Telugu)
 - Fertility: One word in source sentence can give rise to multiple words in target.
- **Semantic ambiguity**: A word can have multiple meanings (polysemy) and many words can have same meaning (synonymy). Translating these types of words are always difficult

History of MT

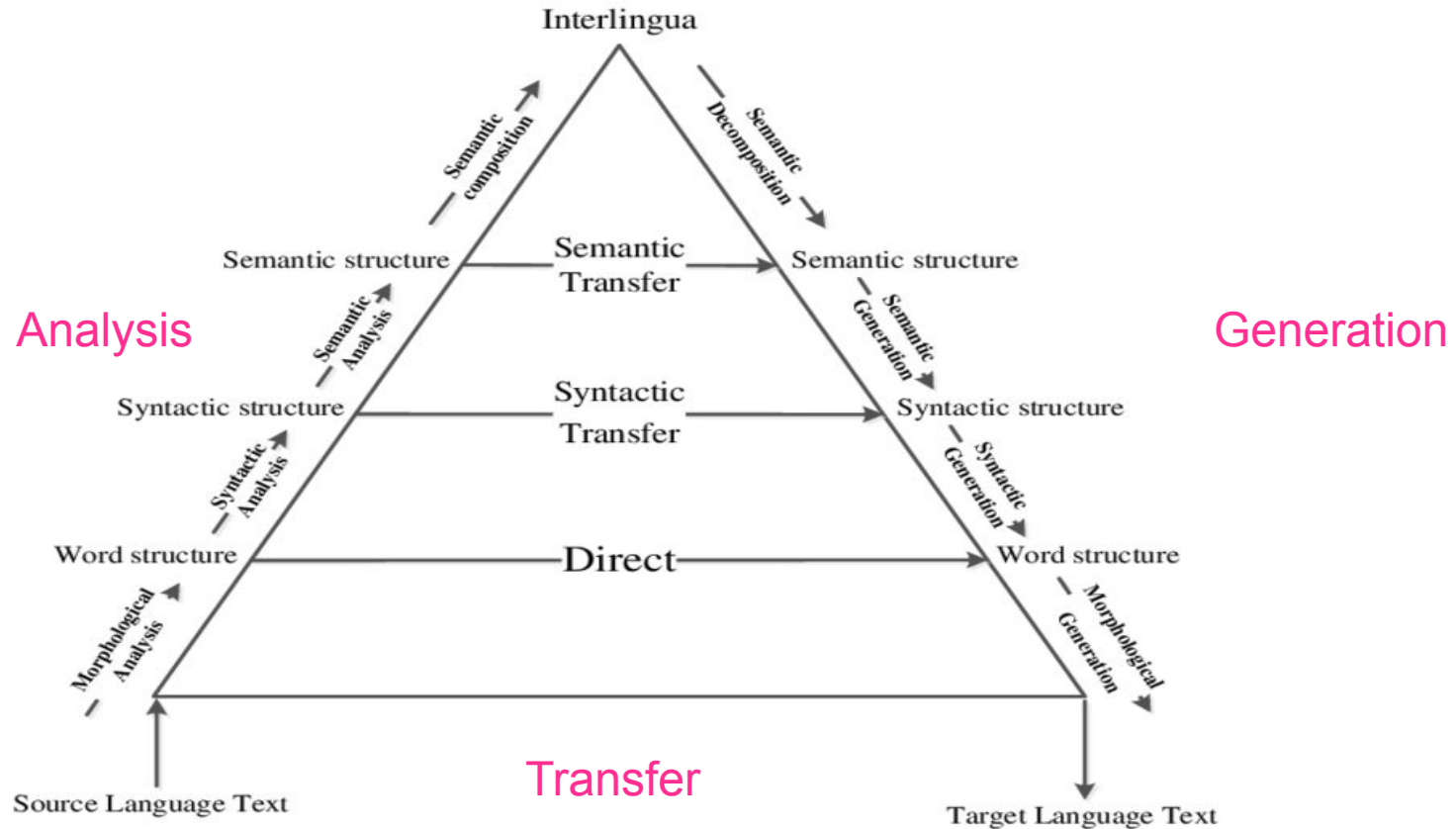
- Since mid 2010s: Neural Network models for MT
 - 2016: New state-of-the-art (NMT)
- Mid 2000s: Phrase-based MT (Google and Moses)
- 1990s: Statistical MT and IBM models
- 1970s: Rule-based systems
- 1947-48: idea of dictionary-based direct translation
- 17th century: Leibniz and Descartes put forward proposals for codes which would relate words between languages
- 196 BCE: Rosetta stone
 - Ancient Greek and Egyptian



MT Paradigm



Rule-based MT: The Vauquois Triangle



Rule-based MT

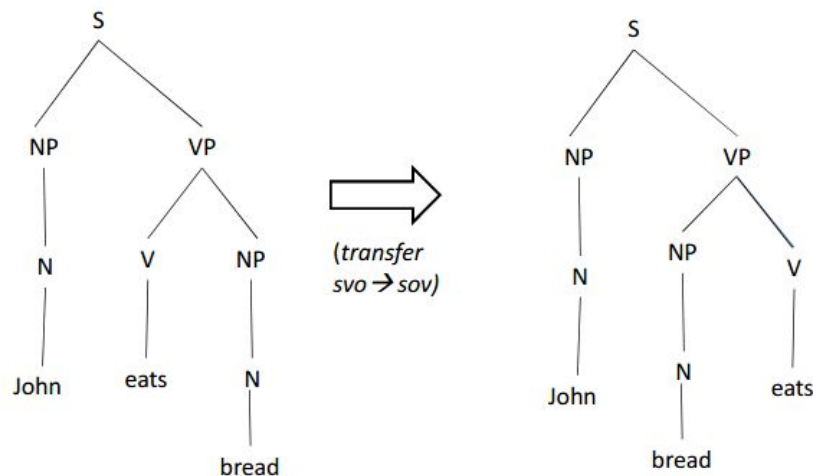
- Create an intermediate representation that captures the meaning of source sentence
- Intermediate representation → Dependent on the pair of language
 - Analysis: Syntactically parse the source sentence
 - Transfer: Rules to convert source parse tree into parse tree for target language
 - Generation: Generate target sentence from that parse tree

Source (English) : “ John eats bread. ”

Target (Hindi) : “ जॉन रोटी खाता है। ”

Rule: Reordering the words

- Word order in source: SVO
- Word order in target: SOV



Event: Hiting,
Agent: John,
Tense: Past,
Polarity: Negative,
Theme: [

Rule-based MT: Interlingua

- Intermediate representation → Independent of the source language
 - Analysis: Encode the source sentence into meaningful representation
 - No Transfer
 - Generation: Generate target sentence from meaning

John hit Sam with a stick.	Event: Hiting, Agent: John, Tense: Past, Polarity: Negative, Theme: [Sam, Attribute: [Object: Stick]]	“ जॉन ने सैम को डंडे से मारा।” (John ne Sam ko dande se maara)
POS tagging Parsing Clause marking Text simplification Named Entity Recognition (NER) Word Sense Disambiguation (WSD)		Lexeme selection Case identification Morphology generation Function word insertion Syntax planning

Analysis

Generation

Example-based MT

- **Input:** He buys a book on international politics

Corpus matches:

He buys a toy

I read a book on international politics

वह एक खिलौना खरीदता है

मैंने अंतरराष्ट्रीय राजनीति पर एक किताब पढ़ी

- **Translation:** वह अंतरराष्ट्रीय राजनीति पर एक किताब खरीदता है

Statistical MT (SMT)

$$\begin{aligned} t^* &= \operatorname{argmax}_t P(t | s) \\ &= \operatorname{argmax}_t \boxed{P(s | t)} \boxed{P(t)} \end{aligned}$$

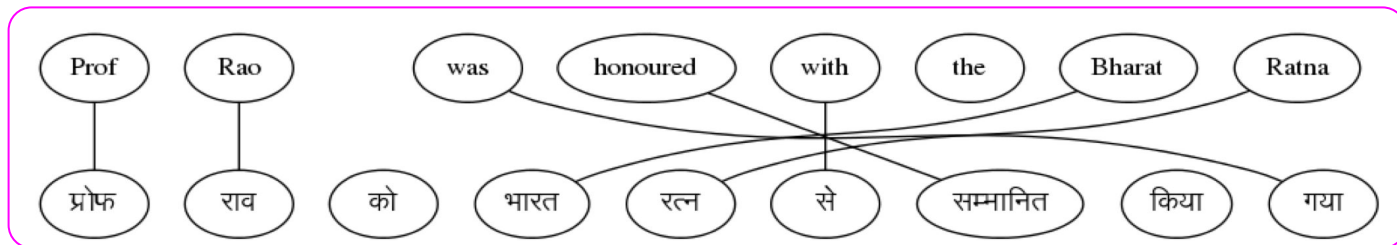
Translation Model Language Model

- **Translation model:** To learn sentence-level translation probabilities, we need to learn word-level translation probability.

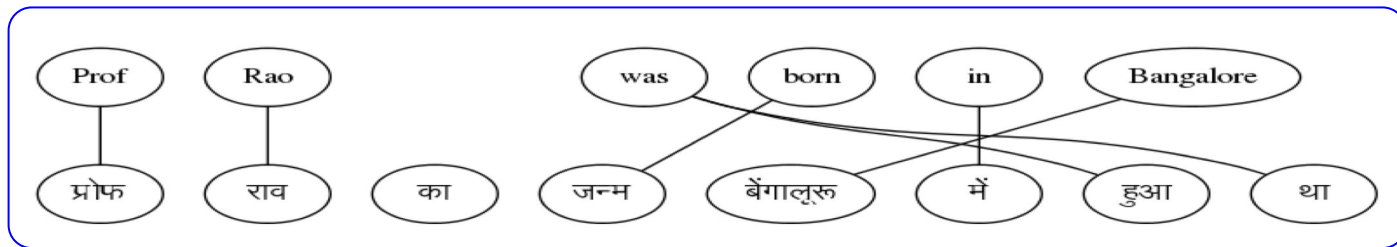
Translation Model

Word-Alignment task

Sentence 1:



Sentence 2:



If we know these alignments, computation of $P(s | t)$ is straight-forward.

$$P(w_s | w_t) = \#(w_s, w_t) / \#(w_t)$$

$$P(\text{Prof} | \text{प्रोफ़}) = \#(\text{Prof}, \text{प्रोफ़}) / \#(*, \text{प्रोफ़}) = 1$$

Translation Model

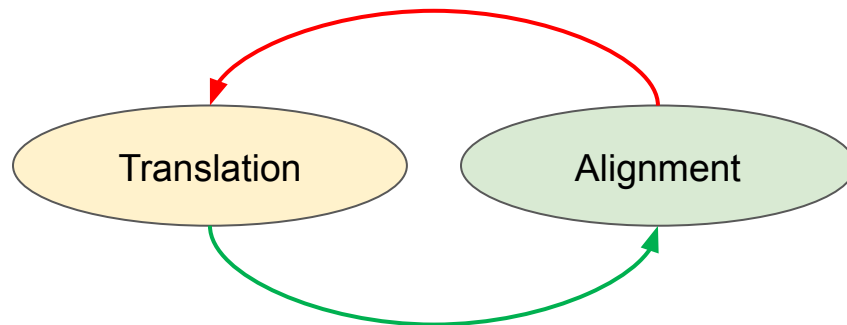
Alignment Model

$$P(s, a | t) = P(a) \prod_{i=1}^m P(s_i | t_{a_i})$$

How to compute the alignment model?

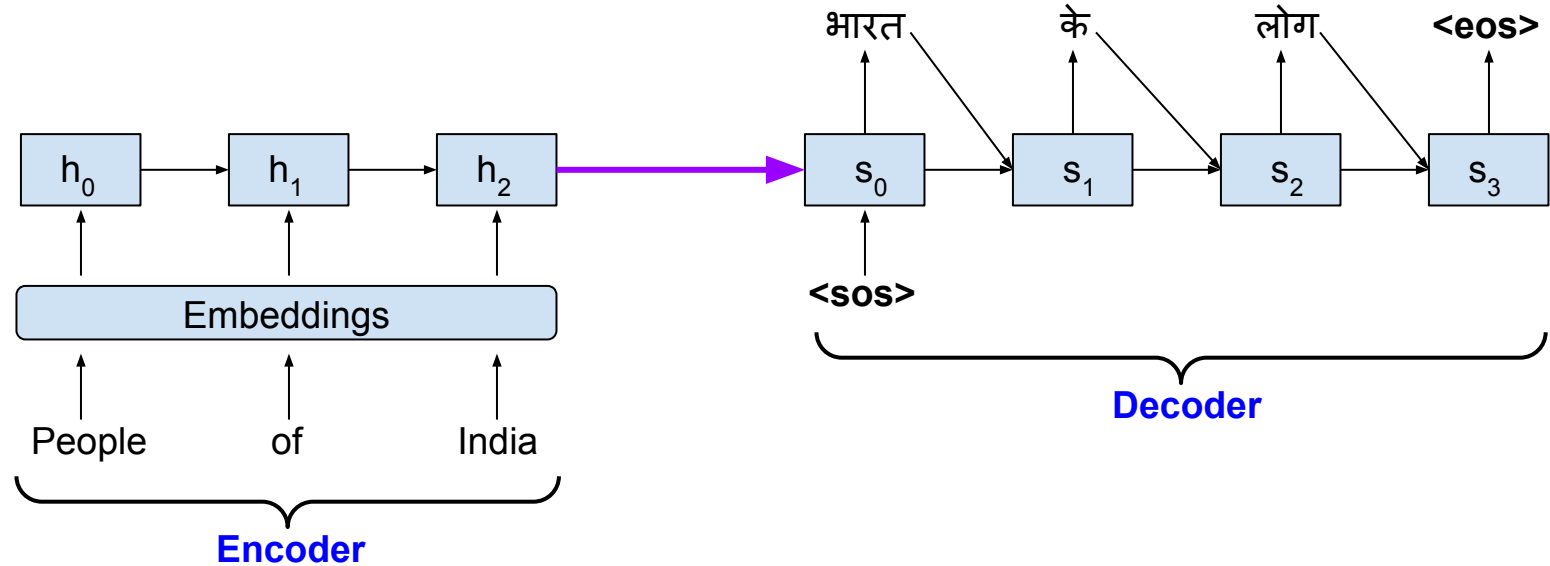
We can find the best alignment only if we know the word translation probabilities

$$a^* = \operatorname{argmax}_a \prod_{i=1}^m P(s_i | t_{a_i})$$

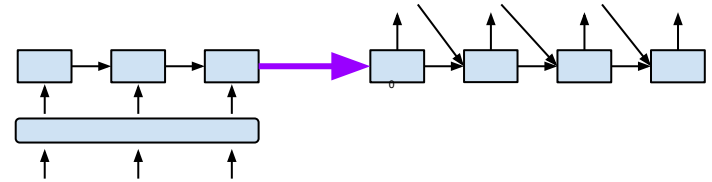


Solution:
Expectation Maximization (EM) Algorithm

Neural MT: Encode-Decode paradigm



Encode-Decode paradigm



- Problem

- Decoding at each timestep is dependant on the sentence vector.

- Solution

- To improve the quality of predictions we can
 - Improve the quality of sentence representation **‘OR’**
 - Present the source sentence representation for prediction at each time step. **‘OR’**
 - Present the *RELEVANT* source sentence representation for prediction at each time step.
 - *Encode - Attend - Decode* (Attention mechanism)

Neural Machine Translation by Jointly Learning to Align and Translate

Bahadanu et al., (ICLR-2015) <https://arxiv.org/pdf/1409.0473.pdf>

Attention

Attention based Enc-Dec

For generation of i_{th} output character:

C_i : context vector

a_{ij} : annotation weight for the j_{th} vector

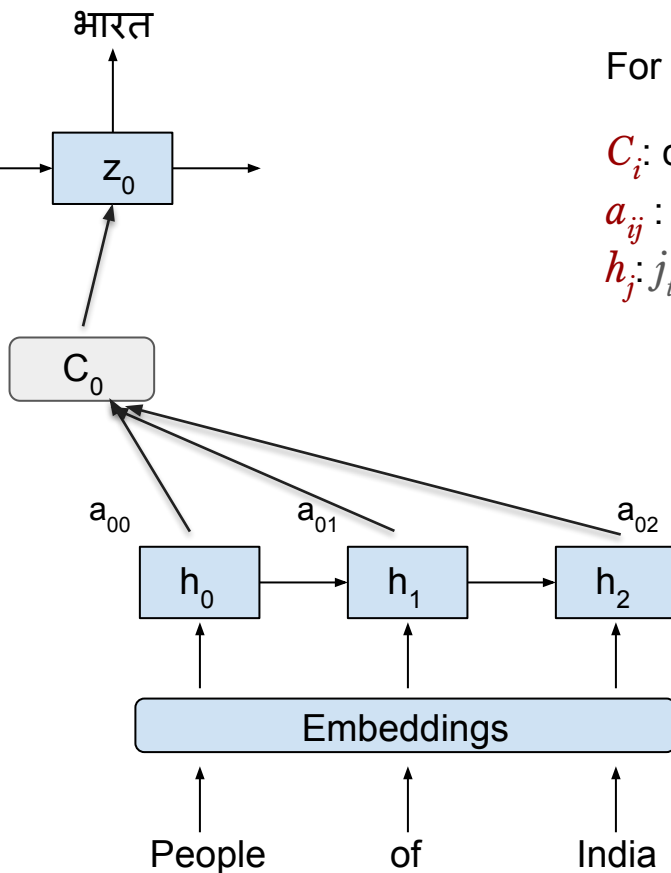
h_j : j_{th} annotation vector

$$c_i = \sum a_{ij} h_j$$

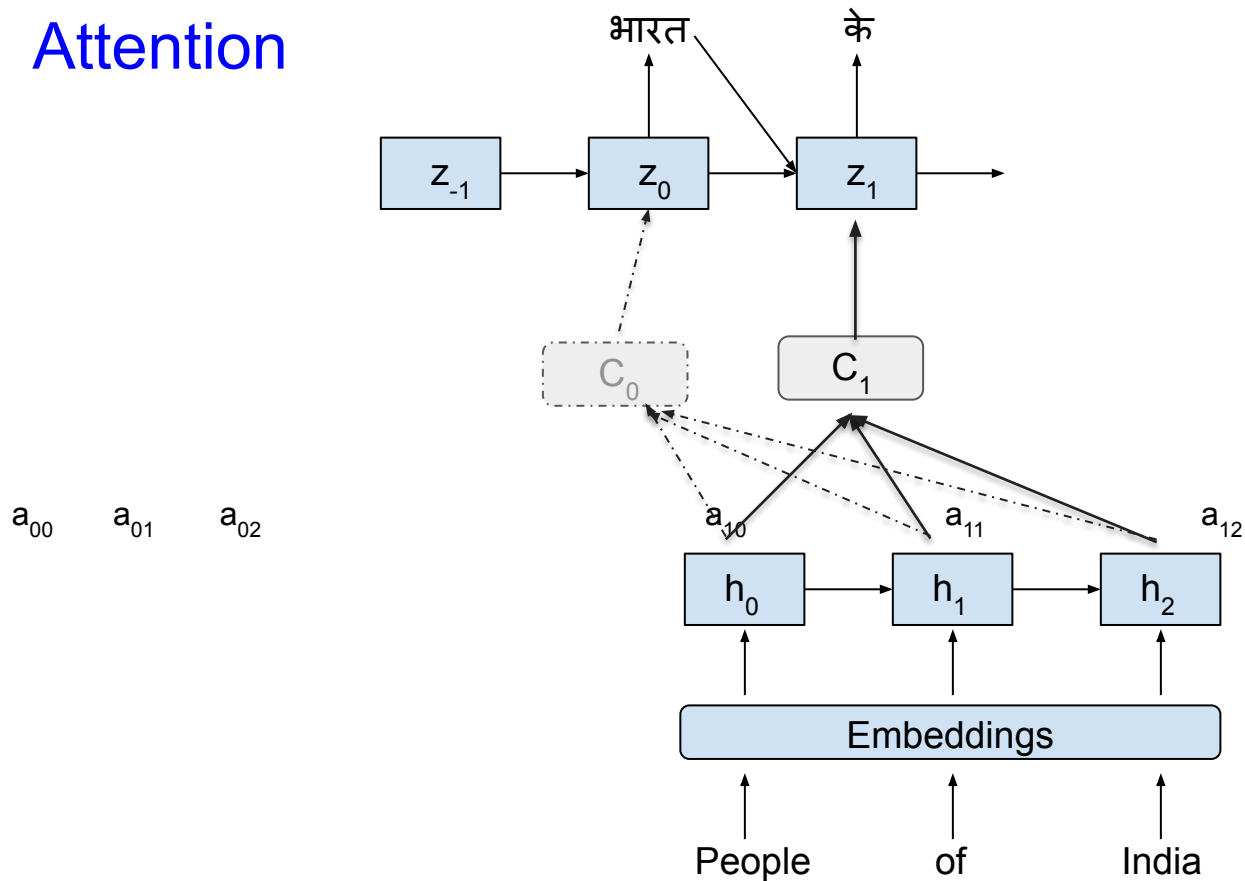
$$a_{ij} = \frac{\exp(e_{ij})}{\sum \exp(e_{ik})}$$

$$e_{ij} = a(z_{i-1}, h_j)$$

$$z_i = f(z_{i-1}, y_{i-1}, c_i)$$



Attention



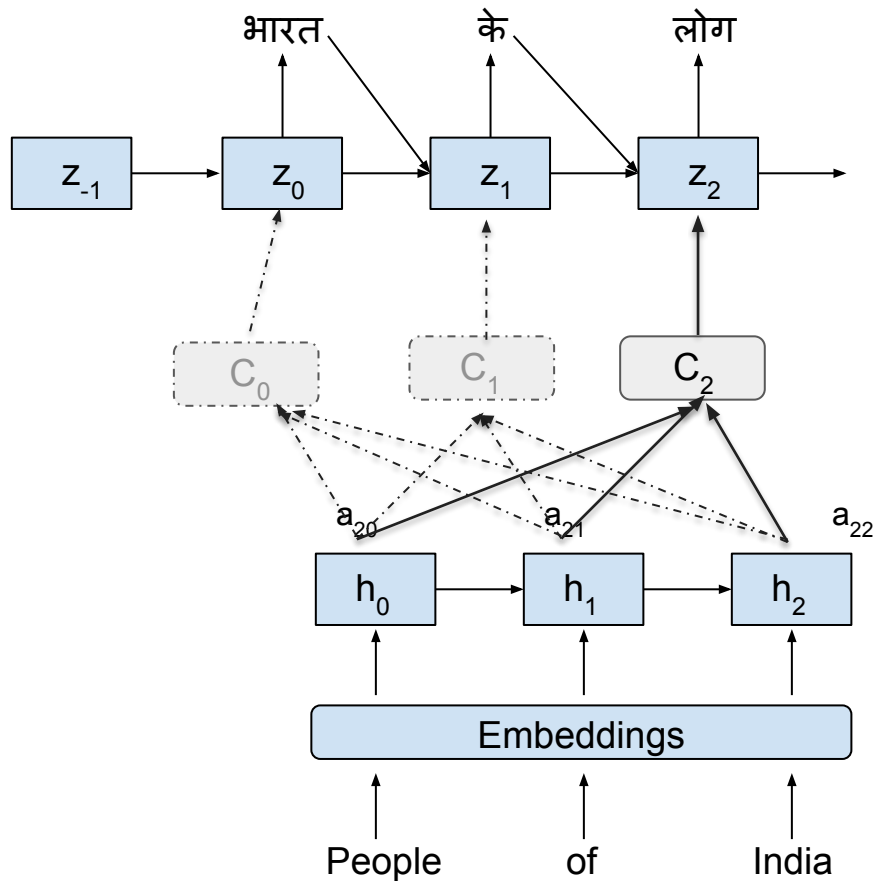
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Attention



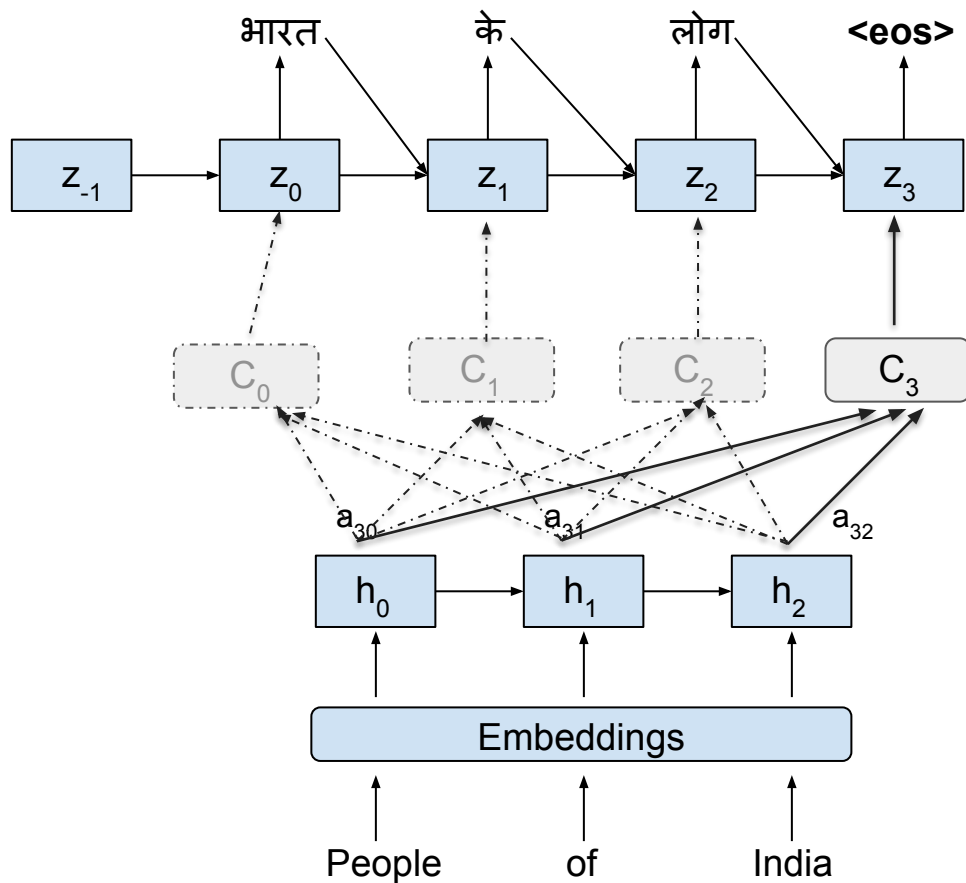
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Attention



a_{00}	a_{01}	a_{02}
a_{10}	a_{11}	a_{12}
a_{20}	a_{21}	a_{22}

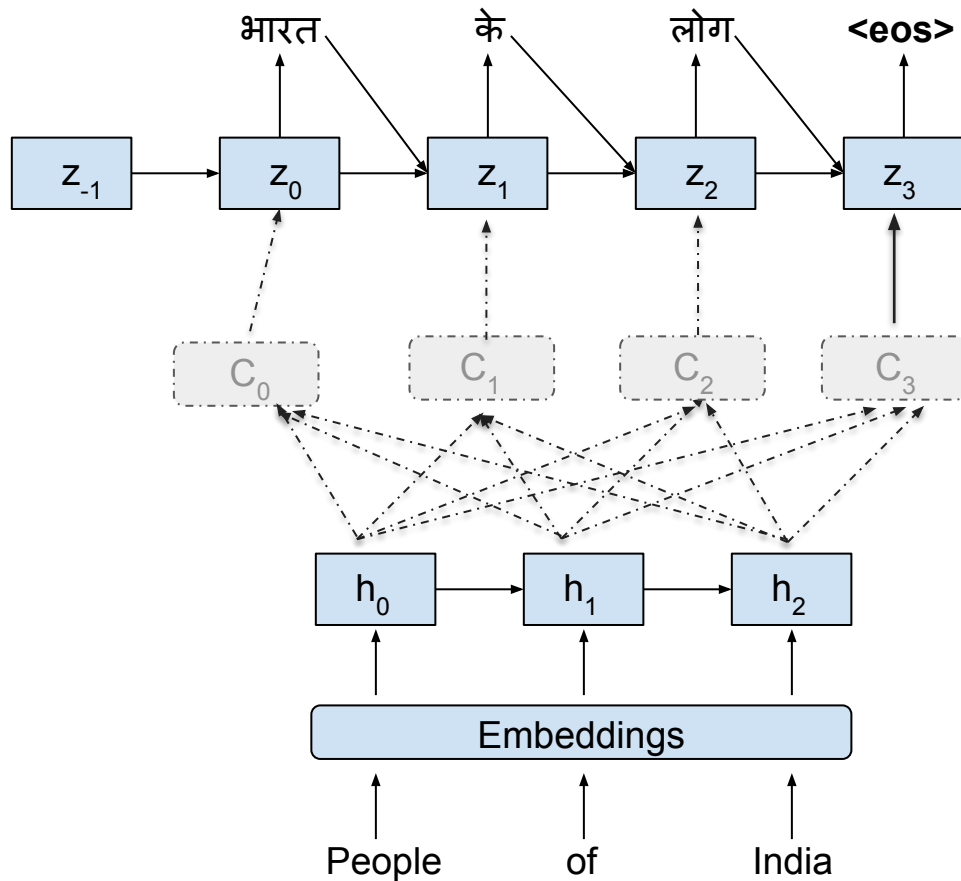
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Attention

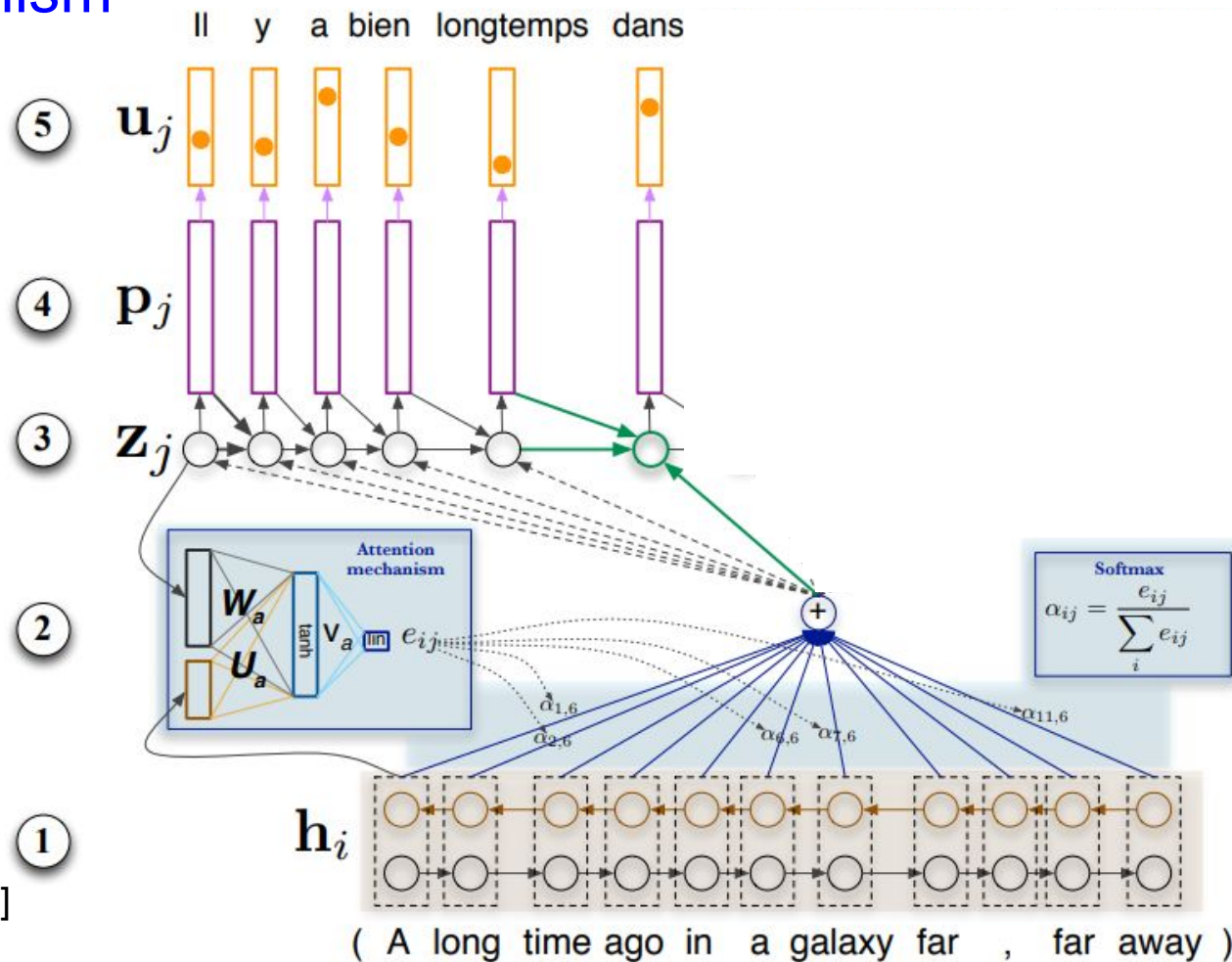


$$\begin{bmatrix} a_{00} & a_{01} & a_{02} \\ a_{10} & a_{11} & a_{12} \\ a_{20} & a_{21} & a_{22} \\ a_{30} & a_{31} & a_{32} \end{bmatrix}$$

Attention weight matrix

Attention Mechanism

1. Bi-RNN Encoder
2. Attention
3. RNN Decoder
4. Output Embeddings
5. Output probabilities



[García-Martínez et al., 2016]

[Bahadanu et al., 2015]

Applications of Attention: ABSA

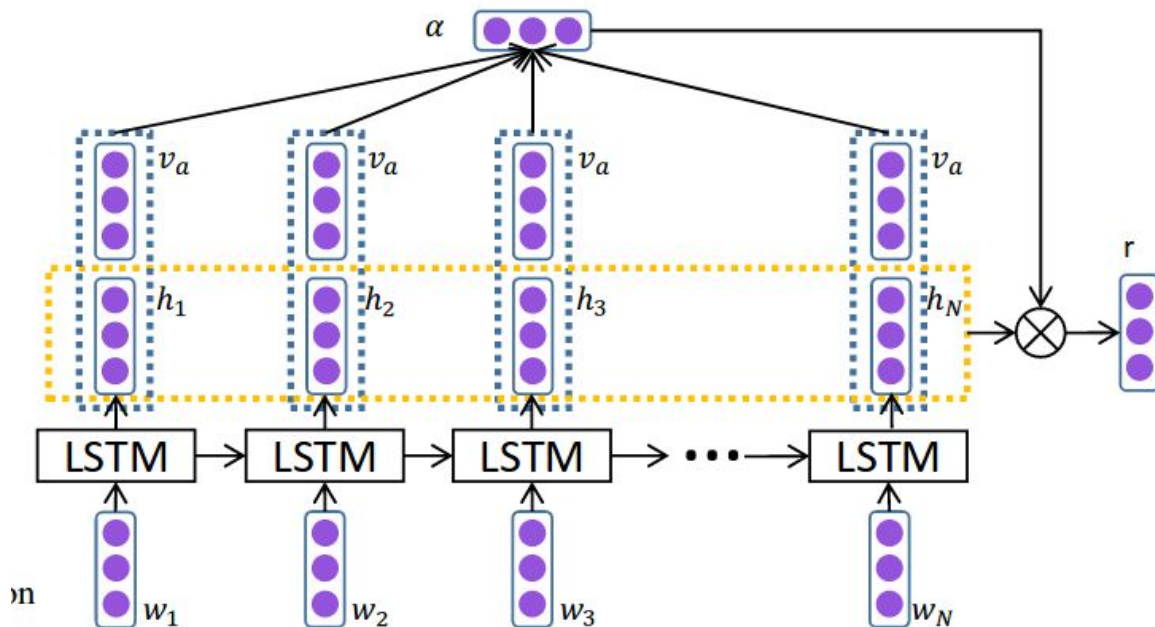
Attention-based LSTM for Aspect-level Sentiment Classification [Wang et al., 2016]

Staffs are not that friendly, but the taste covers all.

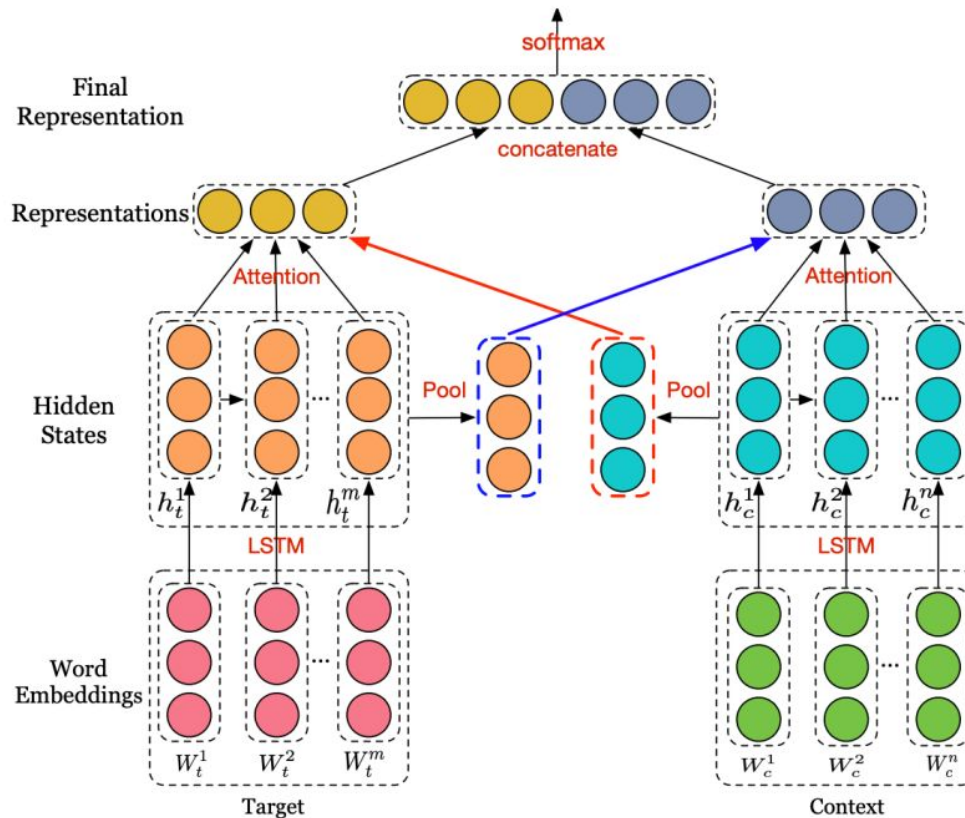
- Aspects / Aspect Category
 - Service** → *negative*
 - Food** → *positive*

Instances and attentions:

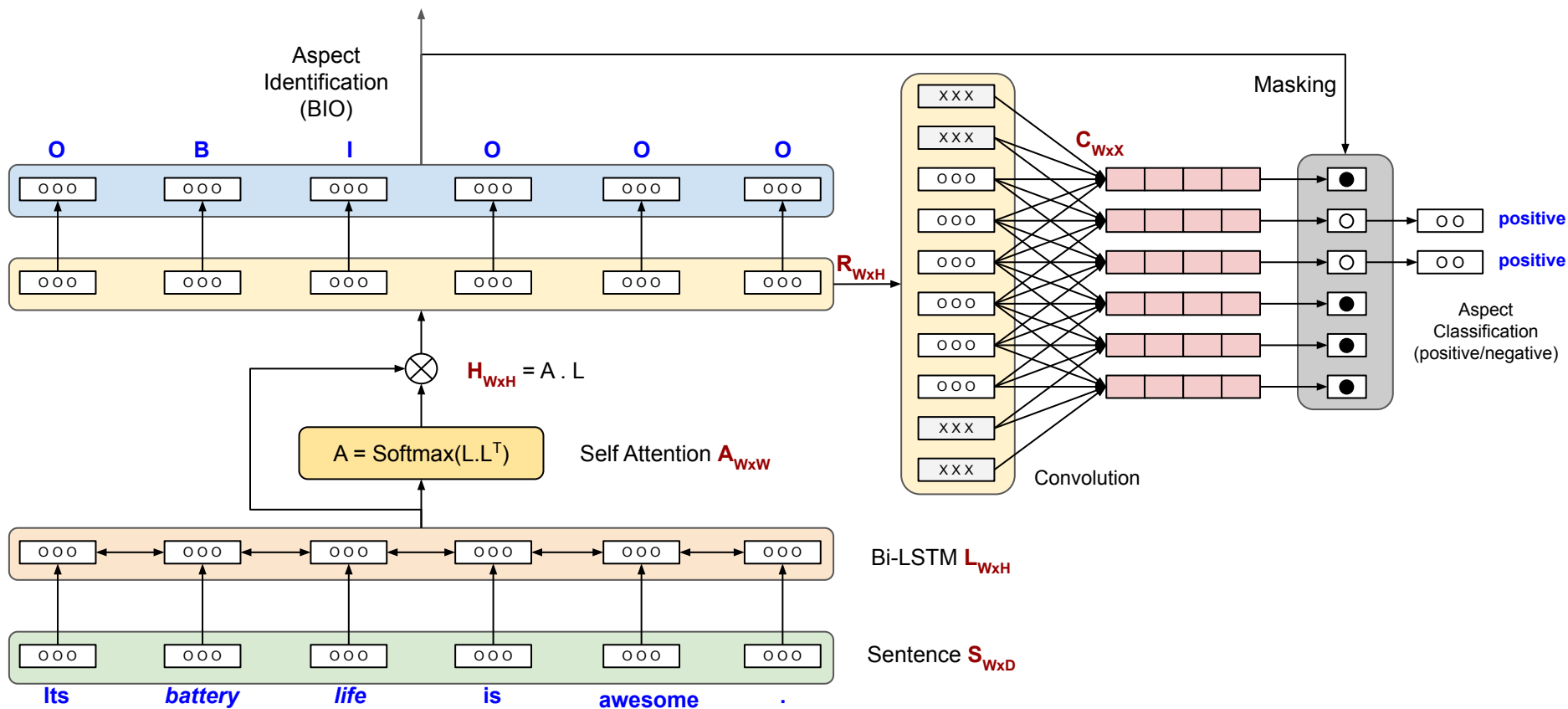
- {Staffs are not that friendly, but the taste covers all, **Service**}*
- {Staffs are not that friendly, but the taste covers all, **Food**}*



Interactive Attention Network - [Ma et al., 2017]



Joint-learning for Aspect-term and sentiment classification [Akhtar et al., 2019]



Types of Attention

- Standard (Soft) attention
- Hard attention - Show, Attend, and Tell (<https://arxiv.org/pdf/1502.03044.pdf>)
- Interactive attention
- Self-attention (Next lecture)
- Quasi attention
(<https://papers.nips.cc/paper/2019/file/16fc18d787294ad5171100e33d05d4e2-Paper.pdf>)
- Gumble attention (<https://arxiv.org/pdf/2103.08862.pdf>)
- Cross-modal/Bi-modal attention (<https://arxiv.org/pdf/1904.04745.pdf>)
- ...

Available MT systems (1/2)

- Supports Indian languages
 - Google Translate (<https://translate.google.com/>) supports Hindi, Bengali, Marathi, Punjabi, Gujarati, Tamil, Telugu, Kannada, Malayalam, Oriya,
 - Microsoft's Bing Translator (<https://www.bing.com/translator>) supports Hindi, Bengali, Marathi, Punjabi, Gujarati, Tamil, Telugu, Kannada, Malayalam
 - Facebook Translate (<http://www.facebook.com/translations>) supports Hindi, Bengali, Marathi, Punjabi, Gujarati, Tamil, Telugu, Kannada, Malayalam, Oriya
 - Amazon Translate (<https://aws.amazon.com/translate/>) supports Hindi, Bengali, Tamil
 - Yandex Translate (<https://translate.yandex.com/>) supports Hindi, Bengali, Marathi, Punjabi, Gujarati, Tamil, Telugu, Kannada, Malayalam

Available MT systems (2/2)

- Does not support Indian languages
 - Systran (<https://www.systransoft.com/>)
 - Lingua Custodia (<https://www.linguacustodia.finance/en/home/>)
 - BabelFish (<https://www.babelfish.com/>)
 - SDL (<https://www.sdl.com/>)
 - Kantan MT (<https://www.kantanmt.com/index.php>)

Available MT APIs for development

- Google Translate api (<https://cloud.google.com/translate/docs/quickstarts>)
- Microsoft Text Translation api
(<https://docs.microsoft.com/en-us/azure/cognitive-services/translator/>)
- Amazon Translate api (<https://docs.aws.amazon.com/translate/latest/dg/what-is.html>)
- IBM Watson Language Translator
(<https://www.ibm.com/watson/services/language-translator/>)
- Systran api (<https://rapidapi.com/systran/api/systran-io-translation-and-nlp>)
- Yandex Translate api
(<https://tech.yandex.com/translate/doc/dg/concepts/api-overview-docpage/>)

Dataset

- Workshop on Machine Translation (WMT) <http://www.statmt.org/wmt20/>
- Workshop of Asian Language Translation (WAT) <http://orchid.kuee.kyoto-u.ac.jp/WAT/>

Parallel corpus available for many language pairs.